

解书照 SHUZHAO XIE

清华大学博士研究生 | 卡内基梅隆大学访问研究助理

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聚焦多模态 Agent 系统与强化学习决策，具备任务分解、工具与技能封装、反馈验证，以及状态 / 动作 / 奖励设计、策略训练与评测经验。结合音视频与三维任务的数据采集和生成，为提升 Coding Agent / LLM 的工具使用与任务执行能力提供支持。

🎓 教育背景

清华大学

◦ 工学博士，计算机科学与技术。导师：王智教授。

中国，深圳

2024.08 – 至今

清华大学

◦ 工程硕士，计算机技术。导师：王智教授。

中国，深圳

2020.06 – 2023.08

北京师范大学

◦ 理学学士，计算机科学与技术。

中国，北京

2016.08 – 2020.06

🖥️ 研究与工作经历

卡内基梅隆大学，CoDimRecon: 多模态 Agent 系统

2026.04 – 2026.10

- 访问研究助理，合作导师：Minchen Li 教授；主导 CoDimRecon，通过多模态理解与 Blender 代码工具调用，将视频转化为可编辑、可仿真的三维场景。
- 设计任务分解与分类型 Agent 工作流，为曲线、曲面和体积组织独立会话与上下文；整合几何先验和生成网格，将物理模型、脚本及示例封装为可复用技能，支持场景构建与仿真初始化。
- 构建执行、验证与修正闭环：结合渲染反馈及机器人仿真交互，诊断几何、运动、数值与材料问题并定向修正；生成场景与渲染数据，为 Agent 的三维建模、编辑与工具使用提供训练和评测素材。

清华大学，Armol / SkyML: 强化学习训练与模型路由

2020.07 – 2023.08

- 面向视觉推理服务选择构建组合强化学习：以图像特征为状态，将连续动作映射为离散服务子集，以预测精度与调用成本共同定义奖励，实现模型路由与组合调用。
- 基于 SAC、经验回放与双 Q 网络开展策略训练，完成 PPO / TD3 对比评测；在相近精度下将推理费用降低 67%。相关成果一作发表于 INFOCOM 2022 / TMM 2024，训练与评测代码已开源。

清华大学，3DGS 约束优化与并行搜索

2023.10 – 2025.10

- 开发 MesonGS/SizeGS/MesonGS++，融合混合整数优化、混合精度量化与 CUDA 并行搜索；搜索最多加速 8×，压缩率超过 34×。

喜尔乐科技（初创），肺音数据采集与识别

2022.07 – 2024.07

- 联合创始人兼算法工程师，参与真实肺音数据采集，为呼吸音分类模型提供数据支持；主导肺部疾病预筛查算法研发。

腾讯，优图实验室，上海

2022.01 – 2022.06

- 模型加速实习生，使用 Faster-RCNN、ONNX、OpenVINO 与 TensorRT。

📄 论文与预印本 (* 共同一作；本人姓名加粗)

- [1] CoDimRecon: Agentic Reconstruction of Sim-Ready 3D Scenes with Deformable Curves, Surfaces, and Volumes
Shuzhao Xie*, Lelin Wang*, Guying Lin, Zhi Wang, Minchen Li. *arXiv preprint*, 2026.
- [2] Cost Effective MLaaS Federation: A Combinatorial Reinforcement Learning Approach
Shuzhao Xie, Yuan Xue, Yifei Zhu, Zhi Wang. *INFOCOM*, 2022.
- [3] SkyML: A MLaaS Federation Design for Multicloud-based Multimedia Analytics
Shuzhao Xie, Yuan Xue, Yifei Zhu, Zhi Wang. *TMM*, 2024.
- [4] MesonGS++: Post-training Compression of 3D Gaussian Splatting with Hyperparameter Searching
Shuzhao Xie*, Junchen Ge*, Weixiang Zhang, Jiahang Liu, Chen Tang, *et al.* *arXiv preprint*, 2026.
- [5] Tail-Aware Post-Training Quantization for 3D Geometry Models
Sicheng Pan*, Chen Tang*, Shuzhao Xie*, Ke Yang, Weixiang Zhang, *et al.* *arXiv preprint*, 2026.
- [6] SizeGS: Size-aware Compression of 3D Gaussian Splatting via Mixed Integer Programming
Shuzhao Xie*, Jiahang Liu*, Weixiang Zhang, Shijia Ge, Sicheng Pan, *et al.* *ACM MM*, 2025. *Best Paper Candidate*.
- [7] SD-GS: Structured Deformable 3D Gaussians for Efficient Dynamic Scene Reconstruction
Wei Yao*, Shuzhao Xie*, Letian Li, Weixiang Zhang, Zhixin Lai, *et al.* *arXiv preprint*, 2025.
- [8] MesonGS: Post-training Compression of 3D Gaussians via Efficient Attribute Transformation
Shuzhao Xie, Weixiang Zhang, Chen Tang, Yunpeng Bai, Rongwei Lu, *et al.* *ECCV*, 2024.
- [9] IGen: Scalable Data Generation for Robot Learning from Open-World Images
Chenghao Gu*, Haolan Kang*, Junchao Lin*, Jinghe Wang, Duo Wu, Shuzhao Xie, *et al.* *CVPR*, 2026.
- [10] Splatwizard: A Benchmark Toolkit for 3D Gaussian Splatting Compression
Xiang Liu, Yimin Zhou, Jinxiang Wang, Yujun Huang, Shuzhao Xie, *et al.* *CVPR Findings*, 2026.
- [11] Treat Domain-Specific Languages as Design Variables in LLM Agents
Letian Li, Wenyuan Jiang, Xin Yang, Shuzhao Xie, Chenghao Gu, *et al.* *NeurIPS, Position Paper Track*, 2026.
- [12] HDSL: A Hierarchical Domain-Specific Language for Structured 3D Indoor Scene Generation and Localized Editing with LLM Agents
Letian Li, Chao Shen, Shuzhao Xie, Chenghao Gu, ZhengXiao He, *et al.* *Asia ACL*, 2026.

- [13] **GeniWorld: A Generalizable Interactive World Model for Robotic Manipulation via Visual Actions**
Chenghao Gu, Hanyang Yu, Jingbo Zhang, Haitao Lin, Wenya Zhang, Jinghe Wang, Hanglei Jin, **Shuzhao Xie**, *et al.*
arXiv preprint, 2026.
- [14] **RoboStream: Weaving Spatio-Temporal Reasoning with Memory in Vision-Language Models for Robotics**
Yuzhi Huang*, Jie Wu*, Weijue Bu*, Ziyi Xiong, Gaoyang Jiang, Ye Li, Kangye Ji, **Shuzhao Xie**, *et al.*
ECCV, 2026.
- [15] **CurvSpec: Adaptive Multi-Curvature Learning for Partial Relevant Video Retrieval**
Zhen Liu, Letian Li, Jinpeng Wang, **Shuzhao Xie**, Yuzhi Huang, *et al.*
ACM MM, 2026.
- [16] **TAU-Bench: From Anomaly Instance Tracking to Fine-Grained Video Anomaly Understanding**
Kepeng Yang*, Dongxuan Liu*, Rongxin Gao*, Zixin Su, Rui Wu, **Shuzhao Xie**, *et al.*
arXiv preprint, 2026.
- [17] **DynTrace: Tracking Dynamic Object Evidence for 4D Spatio-Temporal Reasoning in MLLMs**
Rongxin Gao, Yuzhi Huang, Dongxuan Liu, Chu Li, Zhenye Wang, Jie Wu, **Shuzhao Xie**, *et al.*
arXiv preprint, 2026.
- [18] **Understanding Bias Terms in Neural Representations**
Weixiang Zhang, Boxi Li, **Shuzhao Xie**, Chengwei Ren, Yuan Xue, *et al.*
NeurIPS, 2025.
- [19] **Music-Aligned Holistic 3D Dance Generation via Hierarchical Motion Modeling**
Xiaojie Li, Ronghui Li, Shukai Fang, **Shuzhao Xie**, Xiaoyang Guo, *et al.*
ICCV, 2025.
- [20] **EVOS: Efficient Implicit Neural Training via EVOLutionary Selector**
Weixiang Zhang, **Shuzhao Xie**, Chengwei Ren, Siyi Xie, Chen Tang, *et al.*
CVPR, 2025.
- [21] **Enhancing Implicit Neural Representations via Symmetric Power Transformation**
Weixiang Zhang, **Shuzhao Xie**, Chengwei Ren, Shijia Ge, Mingzi Wang, *et al.*
AAAI, 2025.
- [22] **Expansive Supervision for Neural Radiance Fields**
Weixiang Zhang, Wei Yao, **Shuzhao Xie**, Shijia Ge, Chen Tang, *et al.*
ICME, 2025.
- [23] **Lungmix: A Mixup-Based Strategy for Generalization in Respiratory Sound Classification**
Shijia Ge, Weixiang Zhang, **Shuzhao Xie**, Baixu Yan, Zhi Wang.
ICASSP, 2025.
- [24] **PulmoScan: A Practical Pulmonary Disease Pre-Screening System**
Baixu Yan, Shijia Ge, Meizi Lu, Weixiang Zhang, **Shuzhao Xie**, *et al.*
ICASSP, 2025.
- [25] **TextIR: A Simple Framework for Text-based Editable Image Restoration**
Yunpeng Bai, Cairong Wang, **Shuzhao Xie**, Chao Dong, Chun Yuan, *et al.*
TVCG, 2025.
- [26] **VGGT-DP: Generalizable Robot Control via Vision Foundation Models**
Shijia Ge, Yijun Liu, Yinxin Zhang, **Shuzhao Xie**, Weixiang Zhang, *et al.*
arXiv preprint, 2025.
- [27] **RFQuant: Retraining-free Model Quantization via One-Shot Weight-Coupling Learning**
Chen Tang*, Yuan Meng*, Jiacheng Jiang, **Shuzhao Xie**, Rongwei Lu, *et al.*
CVPR, 2024.
- [28] **A Joint Approach to Local Updating and Gradient Compression for Efficient Asynchronous Federated Learning**
Jiajun Song, Jiajun Luo, Rongwei Lu, **Shuzhao Xie**, Bin Chen, *et al.*
Euro-Par, 2024.
- [29] **WATER-GS: Toward Copyright Protection for 3D Gaussian Splatting via Universal Watermarking**
Yuqi Tan, Xiang Liu, **Shuzhao Xie**, Bin Chen, Shu-Tao Xia, *et al.*
arXiv preprint, 2024.
- [30] **DragScene: Interactive 3D Scene Editing with Single-view Drag Instructions**
Chenghao Gu, Zhenzhe Li, Zhengqi Zhang, Yunpeng Bai, **Shuzhao Xie**, *et al.*
arXiv preprint, 2024.
- [31] **Tuning-Free Visual Customization via View Iterative Self-Attention Control**
Xiaojie Li, Chenghao Gu, **Shuzhao Xie**, Yunpeng Bai, Weixiang Zhang, *et al.*
arXiv preprint, 2024.

🏆 荣誉与奖项

- ACM Multimedia 最佳论文候选（前 0.5%）。2025
- ACM ICPC/CCPC 区域赛铜牌（×2）。2018, 2019

🔧 核心能力

- **Agent 系统**: 多模态工具调用、任务分解、上下文组织、技能封装与复用、代码执行、反馈验证与失败修正。
- **强化学习**: SAC、状态 / 动作 / 奖励设计、组合动作空间、经验回放、双 Q 网络、PPO / TD3 对比评测。
- **工程能力**: Python、C/C++、CUDA、Bash、PyTorch、Blender、UIPC、混合整数优化。